# CSC163 Computing, Ethics, and Society
# Fall 2026 Midterm Exam

Student Name:_____ ID：_____

*Chapters 1–5 | Time allowed: 60 minutes | Total: 100 points | A calculator is allowed.*

| Total pts | / 100 |
|---|---|

Note: For Q2–Q6, **it is recommended that you provide a short explanation for each answer key.** If your answer is correct, you will get the full point without the explanations. But in case your answer is incorrect, the explanations may earn you some partial credit.

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 |
|---|---|---|---|---|---|
| /30 | /14 | /14 | /14 | /14 | /14 |

**Q1 (30 pts) Multiple-choice questions (1 pt each): enter your answer keys here:**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

For the following multiple-choice questions, each question has exactly one correct answer key. If multiple choices are correct, choose the option “All of the above”. Fill in the answer keys in the table above. **(Answer keys written in the question area will not be counted.)**

**Chapter 1: AI Risk Overview (Q1–6)**

1. What are the four major sources of AI risk?
   A. Hardware, software, networks, and data
   B. Malicious use, AI races, organizational risks, and rogue AIs
   C. Bugs, viruses, spam, and phishing
   D. Startups, universities, militaries, and regulators

2. Which risk from AI-enabled cyberattacks is highlighted in the presentation?
   A. AIs could increase the accessibility, success rate, scale, speed, stealth, and potency of cyberattacks
   B. Cyberattacks can no longer target infrastructure
   C. Attribution of attacks becomes trivially easy
   D. Cyberattacks are unrelated to the risk of war

3. What concern does the presentation raise about surveillance and power concentration?
   A. AI makes information processing impossible to centralize
   B. Corporations and states could misuse AI to concentrate power and sustain totalitarian regimes
   C. Automated states depend more heavily on mortal human dictators
   D. Power concentration is limited to individuals only

4. Which measures are listed as ways to reduce malicious use of AI?
   A. Removing all legal liability from developers
   B. Making AI systems easier to jailbreak
   C. Structured access, legal liability for developers, jailbreak resistance, and defensive technologies such as biodefense and anomaly detection
   D. Abandoning anomaly detection research

5. What are AI racing dynamics as described in the slides?
   A. Risks from competition between militaries, corporations, or others to develop increasingly powerful AI systems
   B. A benchmark for model inference speed
   C. A hardware manufacturing standard
   D. A method for labeling training data

6. What is proxy gaming?
   A. Optimizing an approximate goal until it no longer corresponds to the true goal
   B. Training a model with no objective at all
   C. Testing a model only on its training set
   D. Replacing neural networks with lookup tables

**Chapter 2: AI Fundamentals (Q7–12)**

7. Which definition of AI was adopted in the presentation?
   A. Computer systems performing tasks typically associated with intelligence such as problem solving, decision making, and forecasting
   B. Any electronic spreadsheet
   C. Only humanoid robots
   D. A database with no predictive capability

8. What does generative modeling do according to the slides?
   A. Deletes all training examples
   B. Predicts only binary labels
   C. Prevents new samples
   D. Creates new data that resembles the training data

9. What is the main goal of unsupervised learning in the slides?
   A. Follow explicit output labels
   B. Discover patterns and relationships in unlabeled data
   C. Maximize a reward through actions
   D. Compile source code

10. How does reinforcement learning obtain its learning signals?
   A. Only from static labels
   B. Only from clustering
   C. From an agent’s interaction with an environment and the resulting rewards
   D. From manual rules alone

11. Which advantage of deep learning is highlighted as “performance” in the slides?
   A. Guarantee of zero error in all tasks
   B. Superhuman performance in some domains such as image recognition, chess, and Go
   C. Requirement for no data
   D. Perfect transparency in all models

12. Which statement captures the “scalability” property of deep learning mentioned in the presentation?
   A. Performance usually gets worse with more data
   B. Performance quickly saturates and cannot improve
   C. Deep-learning models are highly scalable and performance often improves with more data rather than tapering off
   D. Deep-learning models cannot use large datasets

**Chapter 3: Single-Agent Safety (Q13–18)**

13. Which statement best describes the scope of ML safety in relation to AI safety?
   A. ML safety covers every social and political consequence of AI
   B. ML safety is a small subset of the broader sociotechnical problem of AI safety
   C. ML safety concerns only the speed of model training
   D. ML safety excludes robustness and monitoring

14. In the lecture's safety framework, what is the primary goal of robustness?
   A. Withstand hazards
   B. Identify hazards
   C. Reduce inherent model hazards
   D. Reduce risks beyond individual models

15. What is an adversarial distortion designed to do to an undefended neural network?
   A. Compress its parameters
   B. Improve its calibration
   C. Remove its training data
   D. Cause it to make a mistake

16. When an adversary cannot access model parameters and observes only model outputs, the model is treated as what?
   A. A transparent model
   B. A white-box model
   C. A black-box model
   D. An unlearned model

17. What is the defining characteristic of a Trojan in a machine-learning model?
    A. The model has a large number of parameters
    B. The model cannot be evaluated
    C. Hidden functionality triggers a sudden dangerous behavior change
    D. The model always fails on clean inputs

18. Which three areas are emphasized under monitoring in the lecture?
    A. Compression, routing, and encryption
    B. Planning, control, and actuation
    C. Evaluations, anomaly detection, and transparency
    D. Watermarking, malware creation, and data collection

**Chapter 4: Safety Engineering (Q19–24)**

19. In the deck's terminology, which term describes a hazard with malicious or adversarial intent?
    A. Threat
    B. Failure mode
    C. Exposure
    D. Vulnerability

20. The detailed disaster risk equation expands the classic risk equation into which four factors?
    A. Probability, severity, redundancy, and transparency
    B. Hazard, threat, failure mode, and fail-safe
    C. Reliability, nines, tail, and drift
    D. Probability, severity, exposure, and vulnerability

21. According to the Swiss Cheese Model, safety can be improved by:
    A. Reducing the size and number of holes and increasing the number of slices
    B. Aligning the holes in every slice so that hazards pass through quickly
    C. Enlarging the holes so that failures are easier to notice and fix
    D. Removing slices so that the overall system stays as simple as possible

22. How does Fault Tree Analysis proceed?
    A. Start from a single component and predict every outcome it could eventually cause
    B. Average the failure rates of every component to estimate overall system safety
    C. Start from a specific negative outcome and work backward to its possible causes
    D. Assume accidents happen only when one component fails entirely on its own

23. What is a tail event?
    A. An outcome close to the average
    B. An event that is always anticipated by experts
    C. A high-probability event with small consequences
    D. A low-probability but high-impact occurrence

24. How does the deck define a black swan?
    A. Any event whose estimated probability is below one percent in a given year
    B. A well-understood and frequently observed risk that organizations deliberately accept
    C. A tail event that is largely unanticipated, because it is an unknown unknown
    D. A frequent event whose timing and size can be fully predicted from past data

**Chapter 5: Complex Systems (Q25–30)**

25. What does the reductionist paradigm claim?
    A. Whole systems always exhibit properties that no analysis of their parts can ever explain
    B. Only statistical averages over many systems are meaningful in science
    C. Systems can be fully understood and described by understanding their parts
    D. Systems should never be studied by analyzing their individual components

26. Which phrase captures why emergent features often "turn on" as a system scales up?
    A. Smaller is stronger: systems become safer as they shrink in size
    B. More is different: a quantitative change produces a qualitative one
    C. More is the same: adding components never changes behavior
    D. Faster is safer: higher speed removes emergent behavior altogether

27. What does distributed functionality mean for neural networks?
    A. Nodes or weights do not map to single concepts, so the network is harder to understand but possibly more robust
    B. The network's functions are neatly divided up between separate subsystems that can be studied alone
    C. Each neuron stores exactly one human-readable concept, so the network is easy to inspect
    D. The network cannot be trained on large datasets without losing most of its accuracy

28. What is "risk compensation" in the discussion of adaptive behavior?
    A. Developers refuse to take any risk at all once a system becomes more reliable
    B. Insurance companies automatically pay for any AI accident that occurs in deployment
    C. Systems always become safer as soon as their reliability is improved by engineers
    D. Developers self-organize to their risk tolerance, rendering reliability advances ineffective

29. In Lesson 2 (subgoals can supersede original goals), what does modularization refer to?
    A. A goal is broken into subgoals that may not be pursued in ways that fulfill the original goal
    B. A goal is handed to another agent whose own goals may clash with the original one
    C. A system is rebuilt from scratch every time that it is scaled up in size
    D. Several agents vote on a single shared goal before any of them acts on it

30. What does Lesson 3 say about scaling up a safe system?
    A. A safe system always remains safe when it is scaled up to a much larger size
    B. Scaling up only improves a system's existing abilities and changes nothing else
    C. Scaling up removes emergent capabilities that were present at smaller scales before
    D. A safe system is not necessarily safe when scaled up, because new behaviors can appear

## Q2. (14 pts) Chapter 1: AI Risk Overview

**(a) (8 pts)** Classify each scenario by the AI risk source it illustrates best: malicious use, AI race, organizational risks, or rogue AIs. (2 pts each)

| Scenario | Risk source |
|---|---|
| 1. A criminal group uses an AI model to write personalized phishing messages at scale | |
| 2. Two companies skip safety testing to release their chatbot before a rival does | |
| 3. Incident reports about a deployed AI system are ignored because of a weak safety culture, and a costly accident follows | |
| 4. An AI agent that its developers cannot reliably control resists being shut down | |

**(b) (6 pts)** Bioweapon risk depends on the number of people with the skill and access, multiplied by their willingness to build a weapon. Today 5,000 people have the skill and access, and 0.2 % of them are willing.

| | |
|---|---|
| (i) (2 pts) How many potential attackers are there today? | |
| (ii) (2 pts) LLM assistants multiply the number of people with skill and access by 40. How many potential attackers are there now, and by what factor has the risk grown? | |
| (iii) (2 pts) Structured access (a vetted API) blocks 95 % of the people in (ii). How many potential attackers remain? | |

## Q3. (14 pts) Chapter 2: AI Fundamentals

**(a) (6 pts)** For each scenario give the ML task (classification, regression, anomaly detection, sequence modeling, or sequential decision making) and the learning type (supervised, unsupervised / self-supervised, or reinforcement). (2 pts per row; both entries must be correct)

| Scenario | Task | Learning type |
|---|---|---|
| 1. Filter spam using emails that humans labeled as spam / not spam | | |
| 2. Flag a credit-card purchase that looks very unlike the customer's normal spending (no fraud labels) | | |
| 3. A program learns to play a video game by trial and error, using the game score as reward | | |

**(b) (4 pts)** The six steps of the ML pipeline are listed in scrambled order. Write 1 to 6 for the correct order.

| Step | Order (1–6) |
|---|---|
| Training | |
| Deployment | |
| Labeling | |
| Evaluation | |
| Choosing the architecture | |
| Data collection | |

**(c) (4 pts)** Match each description with one type: narrow AI, AGI, human-level AI (HLAI), transformative AI (TAI), or superintelligence (ASI). Each type is used at most once. (2 pts each)

| Description | Type |
|---|---|
| 1. A system defined by its impact on the world, at the level of the industrial revolution | |
| 2. A system that outclasses humans on virtually all intellectual tasks | |

## Q4. (14 pts) Chapter 3: Single-Agent Safety

**(a) (10 pts)** An anomaly detector gives each input a score and flags the input as out-of-distribution (OOD) if its score is greater than the threshold $\tau = 0.6$. OOD is the positive class. For each input, fill in whether it is flagged and its outcome (TP, FP, FN, or TN).

| Input | True class | Score | Flagged at $\tau = 0.6$? | Outcome |
|---|---|---|---|---|
| A | In-dist | 0.12 | | |
| B | OOD | 0.95 | | |
| C | In-dist | 0.62 | | |
| D | OOD | 0.48 | | |
| E | In-dist | 0.66 | | |
| F | OOD | 0.83 | | |
| G | In-dist | 0.35 | | |
| H | OOD | 0.70 | | |

**(b) (4 pts)** Place each technique in one ML-safety research area: robustness, monitoring, control, or systemic safety. Each area is used once. (1 pt each)

| Technique | Research area |
|---|---|
| 1. Anomaly detection for out-of-distribution inputs | |
| 2. Adversarial training | |
| 3. Machine unlearning of hazardous knowledge | |
| 4. ML intrusion detection for cyberdefense | |

## Q5. (14 pts) Chapter 4: Safety Engineering

**(a) (5 pts)** A system makes a decision each time it must react. Its reliability p is the probability that one decision is correct, its number of nines is $k = -\log_{10}(1 - p)$, and the expected number of decisions until the first mistake is $1/(1 - p)$.

(i) (3 pts) Fill in the table (1 pt per row).

| p | k (nines) | Expected decisions to first mistake |
|---|---|---|
| 0.99 | | |
| 0.9999 | | |
| | 5 | |

(ii) (2 pts) A trip needs 500 decisions and p = 0.999. What is the probability that the whole trip has no mistake?

| | |
|---|---|
| (ii) | |

**(b) (5 pts)** Classic risk equation: Risk = Σ P(hazard) × severity(hazard). Three hazards of an AI deployment are listed below (severity in millions of dollars).

(i) (3 pts) Compute the expected annual risk of each hazard and the total (2 pts). Which hazard has the highest risk, and why (1 pt)?

| Hazard | P (per year) | Severity (M$) | Risk (M$ / year) |
|---|---|---|---|
| AI-enabled ransomware | 0.3 | 6 | |
| Disinformation campaign | 0.02 | 50 | |
| Power-grid control failure | 0.001 | 4000 | |
| Total | | | |

| | |
|---|---|
| (i) Highest-risk hazard and why | |

(ii) (2 pts) Safeguard A cuts the ransomware probability by 50 %. Safeguard B halves the severity of a grid failure. Which one reduces the total risk more, and by how much does each reduce it? (1 pt for each reduction)

| | |
|---|---|
| (ii) | |

**(c) (4 pts)** Match each scenario with ONE safe design principle: redundancy, separation of duties, least privilege, fail-safe, antifragility, negative feedback, transparency, or defense in depth. (2 pts each)

| Scenario | Principle |
|---|---|
| 1. A robot arm stops moving when it loses its sensor signal | |
| 2. A system is protected by a firewall, authentication, encryption, and monitoring together | |

## Q6. (14 pts) Chapter 5: Complex Systems

**(a) (6 pts)** Name the hallmark of complex systems illustrated by each example: emergence, non-linearity and feedback, self-organization, criticality, distributed functionality, scalable structure and power laws, or adaptive behavior. (2 pts each)

| Example | Hallmark |
|---|---|
| 1. Thousands of neurons follow simple local rules, yet the brain as a whole can reason, which no single neuron can do | |
| 2. Adding one more grain of rice to a pile occasionally triggers an avalanche of any size | |
| 3. After cars get anti-lock brakes, drivers brake later and follow more closely | |

**(b) (6 pts)** Kleiber's law: metabolic rate ∝ (body mass)^0.75. A calculator is allowed.

| | |
|---|---|
| (i) (2 pts) By what factor does the rate grow when the mass is multiplied by 16? | |
| (ii) (2 pts) By what factor does the rate grow when the mass is multiplied by 10,000? | |
| (iii) (2 pts) A toy scaling law says error ∝ (data)^−0.1. By what factor does the error change when the data grows ×10? ×1,000? | |

**(c) (2 pts)** Classify each task as a puzzle (one findable answer), a problem (several approaches, but it is clear when it is solved), or a wicked problem (social element, no flawless solution, unclear when solved). (1 pt each)

| Task | Type |
|---|---|
| 1. Diagnose why a web server keeps crashing | |
| 2. Reduce political polarization | |

*— End of exam —*